Dear Editor-in-Chief, dear members of the Editorial Board,

We would like to propose a review article for *Reviews of Geophysics* on wildfire in mountain terrain.

Wildfire is becoming more frequent in mountain environments where it has historically been rare, yet fire science has been informed mainly by lowland and Mediterranean regimes. We argue that mountain wildfire is structurally different among several dimensions.

The review compares mountain wildfire with the non-mountain baseline in six domains: hazard, climate and weather, water, forest, risk, and governance. It takes the European Alps as its anchor case, because they hold the densest record of small fires, protective-forest management and post-fire hazards of any mountain system. To separate what is general to mountains from what is specific to the Alps, each finding will be tested against comparable mountain ranges. These ranges will be selected by a reproducible screening of the world's mountain ranges, using open global datasets, against four criteria: climate, elevation, forest type and fire regime. This spatial selection will steer the literature review away from the European Alps.

We believe the topic suits the journal's readership because mountain wildfire is a coupled geophysical problem. It links atmospheric dynamics in complex terrain, snow and catchment hydrology, hillslope geomorphology and remote sensing of the land surface. It also connects three recent *Reviews of Geophysics* syntheses, none of which is centered on fire in mountains:
- the global review of fire trends and drivers under climate change (Jones et al., 2022)
- the review of elevation-dependent climate change in mountains (Pepin et al., 2022)
- the review of geomorphic process chains in high-mountain regions (Mani et al., 2023)

Existing reviews treat mountain fire meteorology (Sharples, 2009; Werth et al., 2016) and satellite fire mapping (Chuvieco et al., 2019; Wooster et al., 2021) separately, and none that is known to us examines how steep terrain jointly shapes fire weather, detectability and consequence. We will write for geoscientists outside fire science, defining fire-specific terms at first use.

We believe this review is also timely. Mountain fire regimes may be at the start of a shift that recent seasons have only begun to reveal. A synthesis now could help the geoscience and hazard communities prepare for the events “on the front foot”, and it could serve as a common reference for a community of practice across mountain regions.

The proposal builds on an extensive synthesis of the Alpine evidence that the author team has already compiled, so the manuscript can be completed within three months of an invitation. The authors are at Eurac Research, Bolzano, Italy, and work on natural hazards and risk, climate, water, forest ecology and management, ecosystem services and governance. Between them, they cover each of the review's domains.

Thank you for considering this proposal. We would be glad to adjust its scope or emphasis in light of the Board's advice.

With kind regards,

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